

COMPANION HANDOUT

Bayesian Estimation from First Principles

STAT S4204 | Statistical Inference

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Probability theory is nothing but common sense reduced to calculation.

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About this handout. This is a student-written companion to the course notes—not an instructor document. It rebuilds, from nothing, the mathematics behind one sentence from Lecture 2: *under squared-error loss, the best estimator is the posterior mean*. Every term is defined before it is used, every derivation shows every algebraic step, and every step says why it is being taken. No statistics background is assumed; the only prerequisites are single-variable calculus (derivatives, integrals, integration by parts) and patience. Cross-references like 2.9.3 point at numbered objects in the course notes (lecture 2, section 9, object 3); references like B.5.1 point within this handout. A web edition lives on the course site, identical in content, with the figures replaced by live laboratories—the normal–normal update of §B.8 as the centerpiece—so the machinery can be steered by hand while reading.

B.1 The Problem: Guessing a Number You Cannot See

Concept B.1.1

The estimation problem

A new treatment cures some fraction of the patients who receive it. Call that fraction θ : if $\theta = 0.7$, then in the long run 70% of treated patients are cured. Nobody knows θ . It is not random—it is a fixed fact about the treatment—but it is *hidden*: the only way to learn about it is to treat patients and watch what happens. Suppose we treat $n = 10$ patients and $x = 7$ of them are cured. What number should we now report as our best guess of θ ? That question—turn data into a single defensible guess at a hidden number—is called *estimation*, and this handout builds one complete, principled answer to it.

Concept B.1.2

Two sources of information

The obvious answer is the observed cure rate, $7/10 = 0.7$. It is a good answer, and by the end of the handout you will know exactly in what sense. But notice that the data are not the only thing you know. Perhaps treatments of this kind have historically cured between 30% and 60% of patients; perhaps the biology says a cure rate near 1 is implausible. Ten patients is not many, and it seems wasteful to ignore everything except those ten outcomes. The *Bayesian* approach—the subject of this handout—is defined by one move: it expresses *both* sources of information, the data and the background knowledge, in the same language, the language of probability. Once both speak that language, a single theorem combines them optimally.

Remark B.1.3

The route

The handout runs in a straight line, and it helps to see the whole road first. § B.2 builds the probability vocabulary (random variables, distributions, expectation, conditional distributions)—the language everything else is written in. § B.3 makes “estimation” precise and shows why, without background knowledge, the problem gets *stuck*: candidate guessing rules cannot be ranked. § B.4 introduces the Bayesian move—treat the hidden number as random, with a distribution expressing what you believe about it—and derives *Bayes’ theorem*, the rule for updating that belief with data. § B.5 proves the punchline: once beliefs are updated, the best single-number guess under the natural error measure is the *average of the updated belief*. §§ B.6–B.8 then work two complete examples with real numbers, including the ten-patient trial above, and § B.9 asks the honest question—where did the prior belief come from?—before § B.10 hands you the standard vocabulary for further reading.

B.2 The Probability Toolkit, from Zero

Probability is the mathematics of quantities whose values you cannot predict. This section defines the six ideas the rest of the handout uses—random variable, distribution, expectation, variance, conditional distribution, conditional expectation—each with a small computation so the definition is not just words. Readers who have met these before can skim, but should slow down at B.2.7 and B.2.9, which carry the whole later argument.

Definition B.2.1**Random variable; realization**

A *random variable* is a numerical quantity whose value is determined by the outcome of something uncertain—a measurement not yet made. The number of cures among ten patients, before the trial runs, is a random variable; so is tomorrow’s temperature. Random variables are written with capital letters (X, Y); once the uncertainty resolves and a value is observed, that observed number—written in lowercase (x, y)—is called the *realization*. “ $X = 7$ ” is an event that had some probability in advance; “ $x = 7$ ” is what actually happened.

Definition B.2.2**Distribution; pmf; density**

The *distribution* of a random variable is the complete advance description of how likely it is to fall anywhere: the assignment of a probability to every statement of the form “ X lands in the set A .” For a *discrete* random variable (one taking isolated values, like a count) the distribution is carried by the *probability mass function* (pmf)

$$p(x) = P(X = x),$$

the probability of each individual value; probabilities of sets are found by adding: $P(X \in A) = \sum_{x \in A} p(x)$. For a *continuous* random variable (one taking values in a continuum, like a fraction in $[0, 1]$) individual points all have probability zero, and the distribution is instead carried by a *density* $f(x) \geq 0$, with probabilities found by integrating:

$$P(X \in A) = \int_A f(x) dx.^1$$

In both cases the total is 1: $\sum_x p(x) = 1$, $\int f(x) dx = 1$ —*something* must happen.

Definition B.2.3**Expectation**

The *expectation* (or *mean*) of a random variable X is its probability-weighted average value,

$$E\{X\} = \sum_x x p(x) \quad (\text{discrete}), \quad E\{X\} = \int x f(x) dx \quad (\text{continuous}) :$$

each possible value, weighted by how likely it is. Geometrically it is the *balancing point* of the distribution: if the pmf or density were a physical distribution of weight along a rod, $E\{X\}$ is where the rod balances. It is the single number that best summarizes “where the distribution sits”—a claim §B.5 will turn from slogan into theorem.

Example B.2.4

¹A density is not a probability but a probability *rate*: probability per unit length, so that $f(x) dx \approx P(x \leq X \leq x + dx)$ for small dx . Consequently a density can exceed 1 (the uniform density on $[0, \frac{1}{2}]$ is 2 there) and $f(x)$ is emphatically not $P(X = x)$, which is 0. Sums against a pmf and integrals against a density are the same idea with the same rules; the handout states things once and trusts the reader to swap $\sum \leftrightarrow \int$.

Two expectations, fully computed

(a) A random variable Z takes the value 1 with probability θ and 0 with probability $1 - \theta$ (one patient: cured or not; such a Z is called a *Bernoulli*(θ) variable). Its expectation, straight from the definition:

$$E\{Z\} = 1 \cdot P(Z = 1) + 0 \cdot P(Z = 0) = 1 \cdot \theta + 0 \cdot (1 - \theta) = \theta.$$

The mean of a 0-or-1 variable *is* its probability of being 1—the bridge between “probability” and “long-run fraction.” (b) A random variable W takes values 1, 2, 6 with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$. Then

$$E\{W\} = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{3} + 6 \cdot \frac{1}{6} = \frac{1}{2} + \frac{2}{3} + 1 = \frac{3+4+6}{6} = \frac{13}{6} \approx 2.17.$$

Note the balancing-point reading: 2.17 is not a possible value of W ; it is where the weights balance. Expectation also respects sums and constant multiples— $E\{aX + bY\} = aE\{X\} + bE\{Y\}$, always, with no assumptions about how X and Y are related—a property called *linearity* that we will use constantly.²

Definition B.2.5

Variance; standard deviation

The *variance* of X measures its spread: the expected squared distance from its own mean,

$$\text{Var}(X) = E\{(X - E\{X\})^2\}.$$

Squaring makes deviations positive (so left- and right-misses cannot cancel) and charges large deviations disproportionately. Its square root, the *standard deviation*, is the spread in the original units. A useful equivalent form, obtained by expanding the square and using linearity:

$$\text{Var}(X) = E\{X^2 - 2X E\{X\} + E\{X\}^2\} = E\{X^2\} - 2E\{X\}E\{X\} + E\{X\}^2,$$

that is,

$$\text{Var}(X) = E\{X^2\} - E\{X\}^2.$$

For the Bernoulli variable of B.2.4(a): $Z^2 = Z$ (since $0^2 = 0$, $1^2 = 1$), so $E\{Z^2\} = \theta$ and

$$\text{Var}(Z) = \theta - \theta^2 = \theta(1 - \theta),$$

largest at $\theta = \frac{1}{2}$ (a fair coin is the most unpredictable coin) and zero at $\theta = 0$ or 1 (no uncertainty at all).

Definition B.2.6

Joint, marginal, and conditional distributions

²Why linearity is free: in $E\{aX + bY\} = \sum (ax + by)p(x, y)$ the sum splits into $a \sum x p(x, y) + b \sum y p(x, y)$ by ordinary algebra, and the two pieces are $aE\{X\}$ and $bE\{Y\}$. Nothing about the relationship between X and Y was used—only that both means exist.

For a *pair* of random variables (X, Y) , the *joint distribution* assigns a probability to every combination: in the discrete case, $p(x, y) = P(X = x, Y = y)$. From it, the distribution of each variable alone—the *marginal*—is recovered by summing out the other:

$$p_X(x) = \sum_y p(x, y).$$

And the *conditional distribution* of X given that Y has been observed to equal y is the joint, restricted to that observation and rescaled to total 1:

$$p(x | y) = \frac{p(x, y)}{p_Y(y)} \quad (p_Y(y) > 0).$$

Read the formula as a two-step act: *take the slice* of the joint where $Y = y$ (the numerator), then *renormalize* it into a genuine probability distribution by dividing by the slice’s total mass (the denominator). For continuous variables the same formulas hold with densities and integrals.

Example B.2.7

A worked table—and, secretly, the whole handout

A disease affects 10% of a population. A screening test comes back positive for 90% of the sick and (falsely) for 20% of the healthy. Let $D = 1$ mean sick, $T = 1$ mean the test is positive. The joint distribution, built by multiplying each group’s size by its test behavior:

	$T = 1$	$T = 0$	row total
$D = 1$	$0.10 \times 0.90 = 0.09$	$0.10 \times 0.10 = 0.01$	0.10
$D = 0$	$0.90 \times 0.20 = 0.18$	$0.90 \times 0.80 = 0.72$	0.90
column total	0.27	0.73	1.00

The four cells total 1, the row totals are the marginal of D , the column totals the marginal of T . Now suppose your test comes back positive. What is the probability you are sick? Apply B.2.6: slice the $T = 1$ column, then renormalize by its total 0.27:

$$P(D = 1 | T = 1) = \frac{p(1, 1)}{p_T(1)} = \frac{0.09}{0.27} = \frac{1}{3}.$$

Pause on what just happened. *Before* the test, your probability of being sick was 0.10. You received data (a positive test), and your probability *updated* to 0.33—higher, but nowhere near certainty, because false positives are common and the disease is rare. This little table is the entire Bayesian method in miniature: a belief, a data-generating mechanism, an observation, an updated belief. §B.4 will redo this computation with a continuous unknown in place of D and give its parts their proper names.

Definition B.2.8

Conditional expectation

The *conditional expectation* of X given $Y = y$ is the mean of the conditional distribution:

$$E\{X \mid Y = y\} = \sum_x x p(x \mid y)$$

(integral against the conditional density in the continuous case). It is the balancing point of the *slice*: your updated best summary of X after seeing $Y = y$. Note that it is a number for each y —so, before Y is observed, the recipe $y \mapsto E\{X \mid Y = y\}$ is itself a random quantity, written $E\{X \mid Y\}$.

Theorem B.2.9

Iterated expectations

For jointly distributed (X, Y) with $E\{|X|\} < \infty$,

$$E\{X\} = E\{E\{X \mid Y\}\} :$$

the overall mean is the probability-weighted average, over what Y might be, of the slice-by-slice means.

Proof B.2.9

Proof of Iterated Expectations by Regrouping a Double Sum.

- (1) Write out the right side from the definitions. The outer expectation averages the function $y \mapsto E\{X \mid Y = y\}$ against the marginal of Y :

$$E\{E\{X \mid Y\}\} = \sum_y E\{X \mid Y = y\} p_Y(y) = \sum_y \left(\sum_x x p(x \mid y) \right) p_Y(y).$$

- (2) Substitute the definition of the conditional pmf, $p(x \mid y) = p(x, y)/p_Y(y)$:

$$\sum_y \sum_x x \frac{p(x, y)}{p_Y(y)} p_Y(y) = \sum_y \sum_x x p(x, y),$$

the marginal $p_Y(y)$ cancelling exactly—this cancellation is the whole proof.

- (3) Regroup the double sum by x (legal because the sum converges absolutely):

$$\sum_x x \sum_y p(x, y) = \sum_x x p_X(x) = E\{X\},$$

the inner sum being the marginal of X by B.2.6.

■

Uses B.2.6, B.2.8.

Why this theorem matters here: it lets you compute a complicated overall average by *conditioning*—break the world into slices where things are simple, average within each slice, then average the slice-answers. The main theorem of § B.5 is exactly one application of this maneuver.

B.3 Estimation as a Decision, and Why It Gets Stuck

The toolkit is assembled; now aim it at the trial of B.1.1. This section makes four ideas precise—model, estimator, loss, risk—and then demonstrates the problem that motivates everything after: without background knowledge, competing estimators *cannot be ranked*.

Definition B.3.1

The statistical model, and the trial's

A *statistical model* is a family of candidate distributions for the data, indexed by the unknown quantity: for each possible value of θ , a statement of how the data would then behave. The unknown θ is called the *parameter*; the set Θ of its possible values, the *parameter space*. For the trial: each of n patients is independently cured with probability θ , and X counts the cures. A particular sequence with x specified cures and $n - x$ specified failures has probability $\theta^x(1 - \theta)^{n-x}$ (independence multiplies the per-patient probabilities), and there are $\binom{n}{x}$ ways to choose *which* x patients are the cured ones, so

$$f(x | \theta) = P(X = x | \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x = 0, 1, \dots, n,$$

with $\Theta = [0, 1]$. This family is the *binomial* model, written $X \sim \text{Binomial}(n, \theta)$.³

Definition B.3.2

Estimator

An *estimator* is a rule—a function $a(\cdot)$, committed to before the data arrive—that turns any possible data set into a guess: observe $X = x$, report $a(x)$. The *estimator* is the recipe; the *estimate* is the number the recipe produces on your data. Candidate recipes for the trial are cheap to write down: report the observed rate $a_1(x) = x/n$; ignore the data and always report $a_2(x) = \frac{1}{2}$; split the difference, $a_3(x) = (x + 1)/(n + 2)$. Each produces *an* answer on every data set, so producing answers cannot be what makes an estimator good. We need a way to score them.

Definition B.3.3

Loss function; squared-error loss

A *loss function* $L(\theta, a)$ prices the consequence of reporting the guess a when the truth is θ : bigger loss, worse outcome. It is chosen by *us*, when we pose the problem, as the definition of what counts as a miss. This handout uses the standard choice for estimating a number, *squared-error loss*:

$$L(\theta, a) = (\theta - a)^2.$$

³The notation $f(x | \theta)$ —“the pmf of the data *given* the parameter”—is chosen deliberately: from §B.4 on, θ will be treated as a random variable and this really will be a conditional distribution in the sense of B.2.6. The course notes write $f(x; \theta)$ when θ is being treated as a fixed unknown; the two notations name the same function.

Why this and not, say, $|\theta - a|$? Three reasons, in increasing order of importance: squaring is symmetric (over- and under-estimation are punished alike); it charges large misses disproportionately (missing by 0.2 costs four times missing by 0.1), which matches how bad decisions usually scale; and—the deep reason, proved in § B.5—it is the loss under which *averages* are the optimal guesses, making the whole theory computable. Different losses lead to different optimal summaries; squared error leads to means.⁴

Definition B.3.4

Risk

Fix a candidate truth θ and an estimator $a(\cdot)$. Because the data are random, the loss you will suffer is random too; its expectation, computed under the model at that θ , is the *risk*:

$$R(\theta) = E\{L(\theta, a(X))\} = E\{(\theta - a(X))^2\} \quad (X \text{ distributed per } f(\cdot | \theta)).$$

The risk answers: *if the truth were θ , how badly would this recipe do on average?* Note carefully that it is a *function* of θ —one number per candidate truth—because the truth is exactly what we do not know.

Example B.3.5

The risk of the observed rate, computed in full

Take $a_1(X) = X/n$, the observed rate, under the binomial model. First its mean. Writing $X = Z_1 + \cdots + Z_n$ as the sum of the n per-patient Bernoulli indicators and using linearity (B.2.4):

$$E\{X\} = E\{Z_1\} + \cdots + E\{Z_n\} = \underbrace{\theta + \cdots + \theta}_n = n\theta, \quad \text{so} \quad E\left\{\frac{X}{n}\right\} = \frac{n\theta}{n} = \theta.$$

The rate is centered on the truth at *every* θ (such an estimator is called *unbiased*). Its risk is therefore its variance:

$$R_1(\theta) = E\left\{\left(\frac{X}{n} - \theta\right)^2\right\} = E\left\{\left(\frac{X}{n} - E\left\{\frac{X}{n}\right\}\right)^2\right\} = \text{Var}\left(\frac{X}{n}\right),$$

by the definition of variance (B.2.5) with the mean θ substituted. Variances of independent summands add,⁵ and each Bernoulli indicator has variance $\theta(1 - \theta)$ (B.2.5), so

$$\text{Var}(X) = n\theta(1 - \theta) \quad \implies \quad R_1(\theta) = \text{Var}\left(\frac{X}{n}\right) = \frac{\text{Var}(X)}{n^2} = \frac{\theta(1 - \theta)}{n}.$$

Example B.3.6

The stuck comparison

Now score the data-ignoring rule $a_2 \equiv \frac{1}{2}$. Its loss is not random at all—it reports $\frac{1}{2}$ no matter what—so its risk is just

$$R_2(\theta) = \left(\theta - \frac{1}{2}\right)^2.$$

⁴For instance, under absolute-error loss $|\theta - a|$ the optimal guess turns out to be the *median* of one's distribution rather than the mean. The loss function is not a law of nature; it is the posed question, and it selects the answer.

⁵Why: for independent U, V with means μ_U, μ_V , the cross term in $E\{((U - \mu_U) + (V - \mu_V))^2\}$ is $2E\{(U - \mu_U)(V - \mu_V)\} = 2E\{U - \mu_U\}E\{V - \mu_V\} = 0$ —independence lets the expectation of a product factor into a product of expectations, and each factor is zero.

Compare the two at $n = 10$. At $\theta = \frac{1}{2}$:

$$R_1\left(\frac{1}{2}\right) = \frac{\frac{1}{2} \cdot \frac{1}{2}}{10} = 0.025, \quad R_2\left(\frac{1}{2}\right) = 0 :$$

the stubborn rule is *unbeatable*—it happens to be guessing the truth exactly. But at $\theta = 0.9$:

$$R_1(0.9) = \frac{0.9 \times 0.1}{10} = 0.009, \quad R_2(0.9) = (0.9 - 0.5)^2 = 0.16 :$$

now the stubborn rule is nearly eighteen times worse. Neither estimator wins at every θ ; their risk curves *cross*. And this is the general situation, not a quirk of a silly example: reasonable estimators trade performance in one region of Θ for performance in another, and “minimize the risk” is not an instruction anyone can follow, because there is no single risk number to minimize—there is a curve, and curves are not ordered. The problem is stuck. To get unstuck we must be willing to say *which values of θ we care about more*—to put weights on the candidate truths. Doing that with a probability distribution is the Bayesian move, and it is the next section.

B.4 The Prior, the Posterior, and Bayes' Theorem

Definition B.4.1

Prior distribution

A *prior distribution* is a probability distribution $\pi(\theta)$ placed on the parameter itself, expressing what is believed about θ *before* the data are seen: regions of Θ where π is large are values considered plausible, regions where π is small, implausible. Formally this is a change of status: θ is now treated as a *random variable* with density (or pmf) π , so that all of § B.2's machinery—joints, slices, conditional expectations—applies to it. The prior is an *input*, chosen by the analyst as a model of background knowledge, exactly as the binomial model was chosen as a model of the data mechanism. It is not derived from the data and it is not the thing we will optimize; it is a premise.⁶

Definition B.4.2

The joint distribution of parameter and data

Once θ is random, the pair (θ, X) has a joint distribution, and it is built from the two models already on the table, prior first, data mechanism second:

$$\text{joint}(\theta, x) = \pi(\theta) f(x | \theta).$$

Read it generatively: nature draws a truth θ from π , then the experiment draws data x from the model at that truth. (This is the same construction as the table of B.2.7: row weights

⁶Two readings coexist. If θ was literally produced by a random mechanism (a factory turns out batches whose defect rates vary), the prior is that mechanism's distribution, an ordinary frequency statement. More commonly nothing generated θ —the cure rate simply *is*—and the prior is a quantified statement of evidence and judgment: “I regard values near 0.4 as twice as plausible as values near 0.7.” The mathematics downstream is identical under either reading; the interpretation debate is real and is taken up in § B.9.

0.1/0.9 were the prior over D ; the within-row test behavior was the model; cells were prior $\times$ model.)

Definition B.4.3

The marginal distribution of the data

Summing (or integrating) the joint over θ gives the *marginal* distribution of the data,

$$m(x) = \int_{\Theta} \pi(\theta) f(x | \theta) d\theta :$$

how probable the data set x is *overall*, averaged over what the truth might be. In the table it was the column totals. It involves no unknown— θ has been integrated away—and its role in what follows is purely that of a normalizing constant.

Theorem B.4.4

Bayes' theorem

The conditional distribution of the parameter given the observed data—the *posterior distribution*—is

$$\pi(\theta | x) = \frac{\pi(\theta) f(x | \theta)}{m(x)} = \frac{\pi(\theta) f(x | \theta)}{\int_{\Theta} \pi(t) f(x | t) dt}.$$

Proof B.4.4

Proof of Bayes' Theorem by Slicing the Joint.

- (1) By the definition of a conditional distribution (B.2.6), applied to the pair (θ, X) with the roles “condition on the second coordinate”: the conditional of θ given $X = x$ is the joint divided by the marginal of the conditioning variable,

$$\pi(\theta | x) = \frac{\text{joint}(\theta, x)}{m(x)}.$$

- (2) Substitute the joint's factorization $\pi(\theta)f(x | \theta)$ (B.4.2) in the numerator and the integral form of $m(x)$ (B.4.3) in the denominator. (The dummy variable in the denominator is renamed t so it cannot be confused with the free θ in the numerator.)

■

Uses B.2.6, B.4.2, B.4.3.

Remark B.4.5

Reading Bayes' theorem

Three observations, each load-bearing. First, the structure is *slice and renormalize*, nothing more: observing x freezes one slice of the joint; dividing by the slice's total mass $m(x)$ makes it a probability distribution over θ again. The theorem is B.2.6 wearing new labels. Second, look at how the data enter: only through the factor $f(x | \theta)$, read as a function of θ with the observed x plugged in. That function has a name—the *likelihood*—and it scores each candidate truth by how well it explains what was actually seen. Posterior $\propto$ prior $\times$ likelihood: updated belief is old belief reweighted by

explanatory success. Third, the denominator does not involve θ at all. So to identify the posterior it is enough to know the numerator's *shape* in θ ; the constant is whatever makes the total 1. In practice one therefore writes

$$\pi(\theta | x) \propto \pi(\theta) f(x | \theta),$$

drops every factor not involving θ , and tries to *recognize* the resulting shape as a known family. §B.7 runs exactly this play, and the integral in the denominator never gets computed.

Example B.4.6

The table of B.2.7, renamed

You have already done a full Bayesian update. In B.2.7: the prior was $\pi(1) = P(D = 1) = 0.10$; the model was the test's behavior given the disease state, $f(1 | D = 1) = 0.9$ and $f(1 | D = 0) = 0.2$; the observed data was $T = 1$; the marginal was $m(1) = 0.10 \times 0.9 + 0.90 \times 0.2 = 0.27$; and Bayes' theorem gave the posterior

$$\pi(1 | 1) = \frac{\pi(1) f(1 | 1)}{m(1)} = \frac{0.10 \times 0.90}{0.27} = \frac{0.09}{0.27} = \frac{1}{3}.$$

Belief 0.10, data, belief $\frac{1}{3}$. Everything from here on is this computation with a continuous unknown—and then one further question the table never asked: once you have the posterior, *what single number should you report?*

B.5 The Best Guess Under Squared Error

The posterior is a whole distribution over θ —a full account of what is now believed. Often that is the honest thing to report. But estimation demands one number, and now the loss function chooses it. This section proves the two results on which everything rests: with no data, the best single-number guess of a random quantity is its *mean*; with data, the best guess is the mean of the *posterior*. The first result is the engine; the second is one application of iterated expectations to it.

Theorem B.5.1

The best constant guess is the mean

Let θ be a random variable with $E\{\theta^2\} < \infty$. Among all constants a , the expected squared error

$$E\{(\theta - a)^2\} = \text{Var}(\theta) + (E\{\theta\} - a)^2$$

is minimized at exactly one point: $a = E\{\theta\}$, where it equals $\text{Var}(\theta)$.

Proof B.5.1

Proof of the Best Constant by Adding and Subtracting the Mean.

- (1) Write $\mu = E\{\theta\}$ for brevity, and route the error through μ by adding and subtracting it—the step that makes the square split cleanly:

$$\theta - a = (\theta - \mu) + (\mu - a).$$

- (2) Square the two-term sum using $(u + v)^2 = u^2 + 2uv + v^2$:

$$(\theta - a)^2 = (\theta - \mu)^2 + 2(\theta - \mu)(\mu - a) + (\mu - a)^2.$$

- (3) Take expectations term by term (linearity, B.2.4). The first term gives $E\{(\theta - \mu)^2\} = \text{Var}(\theta)$ by definition. In the middle term, $\mu - a$ is a constant and factors out:

$$E\{2(\theta - \mu)(\mu - a)\} = 2(\mu - a) E\{\theta - \mu\} = 2(\mu - a)(E\{\theta\} - \mu) = 2(\mu - a) \cdot 0 = 0 :$$

deviations from one's own mean average to zero, so the cross term dies. The last term is the constant $(\mu - a)^2$, its own expectation.

- (4) What remains is the display: $E\{(\theta - a)^2\} = \text{Var}(\theta) + (\mu - a)^2$. The first piece does not involve a at all; the second is a square, so it is ≥ 0 always, $= 0$ exactly at $a = \mu$, and > 0 elsewhere. The minimizer is $a = \mu$, uniquely. ■

Uses B.2.3, B.2.4, B.2.5.

Remark B.5.2

What the identity says

The proof delivered more than a minimizer; it delivered an exact accounting. Guessing a random quantity with a constant costs you $\text{Var}(\theta)$ *no matter what*—the irreducible price of the quantity's own randomness—plus a penalty $(E\{\theta\} - a)^2$ for centering your guess anywhere but the mean. This is why “balancing point” was the right picture for the expectation in B.2.3: under squared error, the balancing point is provably the best single summary. And note the theorem's scope: *any* random θ , any distribution with a finite second moment. In particular it will apply verbatim to θ under its posterior distribution—which is the next theorem.

Definition B.5.3

Bayes risk

Let $a(\cdot)$ be an estimator. Its *Bayes risk* is its expected squared error when *both* sources of randomness run: the truth θ drawn from the prior, the data X drawn from the model given that truth—that is, (θ, X) carrying the joint distribution of B.4.2:

$$r(a) = E\{(a(X) - \theta)^2\}.$$

Compare B.3.4: the risk froze θ and averaged over data only, producing a curve; the Bayes risk averages over θ as well—the prior supplies precisely the weights over candidate truths that B.3.6 found missing—and produces a *single number* per estimator. Numbers, unlike curves, can be minimized. The estimator with the smallest Bayes risk is called the *Bayes estimator*, and the next theorem identifies it.

Theorem B.5.4

The Bayes estimator is the posterior mean

Assume $E\{\theta^2\} < \infty$. Among all estimators $a(\cdot)$, the Bayes risk $r(a)$ is minimized by reporting, for each possible data set x , the mean of the posterior distribution:

$$a^*(x) = E\{\theta \mid X = x\} = \int_{\Theta} \theta \pi(\theta \mid x) d\theta.$$

Proof B.5.4

Proof of the Posterior-Mean Rule by Conditioning on the Data.

- (1) Iterate the Bayes risk through the data, using iterated expectations (B.2.9) with the conditioning variable X :

$$r(a) = E\{(a(X) - \theta)^2\} = E\left[E\{(a(X) - \theta)^2 \mid X\}\right]:$$

the overall average equals the average, over data sets, of the within-data-set averages.

- (2) Freeze one slice $X = x$ and look at the inner expectation. Inside the slice, $a(x)$ is a fixed *constant*—the recipe's output on that data set—while θ remains random, distributed according to the conditional distribution of θ given $X = x$. And that conditional distribution is, by B.4.4, exactly the posterior $\pi(\cdot \mid x)$. So the inner expectation is the expected squared error of a *constant* guess at a random quantity:

$$E\{(a(x) - \theta)^2 \mid X = x\} = E\{(\theta - c)^2\} \Big|_{c=a(x)} \quad \text{with } \theta \text{ carrying the posterior.}$$

- (3) Apply B.5.1 *to the posterior distribution*—the theorem holds for any distribution, so in particular this one. Within the slice, the inner expectation is minimized, uniquely, by the constant equal to that distribution's mean:

$$c = E\{\theta \mid X = x\},$$

and no other choice of $a(x)$ does as well in this slice.

- (4) The rule $a^*(x) = E\{\theta \mid X = x\}$ therefore minimizes the inner expectation *simultaneously at every x* . Averaging over slices (step (1)) preserves the domination: for any competitor a , each of a 's slice-averages is at least a^* 's, so the overall average is too. Hence $r(a^*) \leq r(a)$ for every estimator a , which is the claim. ■

Uses B.2.9, B.4.4, B.5.1.

Remark B.5.5

The recipe, in words

The theorem compresses the whole method into two verbs: **update, then average**.

1. Model the data mechanism ($f(x | \theta)$) and state your prior (π).
2. Observe x ; form the posterior $\pi(\theta | x) \propto \pi(\theta)f(x | \theta)$ (B.4.4), the updated belief.
3. Report the posterior's mean. That single number minimizes expected squared error among *all* possible estimators, given the premises.

Notice what is chosen and what is given: the prior and the loss are premises, posed with the problem; the theorem then leaves *nothing* to choose—the optimal estimator is determined. The remaining work in any particular problem is computational: actually finding the posterior and its mean. The next three sections do that work twice, in the two settings where it is fully explicit.

B.6 A Toolkit Distribution: the Beta Family

To run the recipe on the trial we need a family of prior distributions for a *fraction*—a random quantity living in $[0, 1]$ —rich enough to express real beliefs and tame enough to compute with. The standard choice is the Beta family. Its normalizing constant involves the Gamma function, so that comes first; both objects repay the ten minutes they take to set up.

Definition B.6.1

The Gamma function

For $s > 0$,

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt.$$

Two facts are all we need. First, the recursion $\Gamma(s+1) = s\Gamma(s)$: integrate by parts with $u = t^s$ (so $du = s t^{s-1} dt$) and $dv = e^{-t} dt$ (so $v = -e^{-t}$):

$$\Gamma(s+1) = \int_0^\infty t^s e^{-t} dt = \left[-t^s e^{-t} \right]_0^\infty + s \int_0^\infty t^{s-1} e^{-t} dt = 0 + s\Gamma(s),$$

the boundary term vanishing at both ends ($t^s e^{-t} \rightarrow 0$ as $t \rightarrow \infty$ because the exponential beats any power; at $t = 0$, $t^s = 0$ for $s > 0$). Second, the anchor $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$. Together they give, for whole numbers, $\Gamma(n) = (n-1)\Gamma(n-1) = \cdots = (n-1)!$: the Gamma function is the factorial, shifted by one and extended to non-integers.

Definition B.6.2

The Beta distribution

For parameters $\alpha > 0$, $\beta > 0$, the *Beta*(α, β) *distribution* is the continuous distribution on $[0, 1]$ with density

$$\pi(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)},$$

the constant $B(\alpha, \beta)$ being exactly what makes the density integrate to 1. The exponents steer the shape: mass shifts toward 1 as α grows and toward 0 as β grows, so (α, β) behave like counts of imagined prior successes and failures—a reading B.7.3 will make exact. One member deserves special mention: $\alpha = \beta = 1$ gives $\pi(\theta) = \theta^0(1-\theta)^0/B(1, 1)$, and since

$B(1, 1) = \Gamma(1)\Gamma(1)/\Gamma(2) = 1 \cdot 1/1 = 1$, this is $\pi(\theta) \equiv 1$: the *uniform* distribution, the flat prior that treats every fraction as equally plausible.

Theorem B.6.3

The mean of a Beta distribution

If $\theta \sim \text{Beta}(\alpha, \beta)$, then

$$E\{\theta\} = \frac{\alpha}{\alpha + \beta}.$$

Proof B.6.3

Proof of the Beta Mean by the Kernel Trick.

- (1) From the definition of expectation, multiplying the density by θ merges the extra θ into the exponent:

$$E\{\theta\} = \int_0^1 \theta \cdot \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)} d\theta = \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1} d\theta.$$

- (2) Recognize the remaining integrand as the *unnormalized* $\text{Beta}(\alpha+1, \beta)$ density. Its integral must therefore be that family's normalizing constant:

$$\int_0^1 \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1} d\theta = B(\alpha+1, \beta).$$

(This maneuver—evaluate an integral by recognizing a known density's shape—is the *kernel trick*, and it is the workhorse of every Bayesian computation to come.)

- (3) So $E\{\theta\} = B(\alpha+1, \beta)/B(\alpha, \beta)$. Expand both via B.6.2 and cancel:

$$E\{\theta\} = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+1)}.$$

- (4) Apply the recursion $\Gamma(s+1) = s\Gamma(s)$ (B.6.1) to each ratio: $\Gamma(\alpha+1)/\Gamma(\alpha) = \alpha$ and $\Gamma(\alpha+\beta)/\Gamma(\alpha+\beta+1) = 1/(\alpha+\beta)$. Hence $E\{\theta\} = \alpha/(\alpha+\beta)$. ■

Uses B.2.3, B.6.1, B.6.2.

B.7 The Cure Rate, Start to Finish

Everything is on the table. Model: $X \sim \text{Binomial}(n, \theta)$ (B.3.1). Prior: to begin, the flat $\text{Beta}(1, 1)$ —"before the trial, no fraction is privileged over any other." Loss: squared error. Data: $x = 7$ cures among $n = 10$ patients. The recipe of B.5.5 says: form the posterior, report its mean.

Example B.7.1

The posterior, by kernel-matching

Bayes' theorem in its proportional form (B.4.5), for a general x and n under the flat prior:

$$\pi(\theta | x) \propto \pi(\theta) f(x | \theta) = 1 \cdot \binom{n}{x} \theta^x (1 - \theta)^{n-x} \propto \theta^x (1 - \theta)^{n-x},$$

the binomial coefficient dropping because it does not involve θ —it is data-only, hence absorbed into the normalizing constant. Now *match the kernel*: rewrite the exponents in Beta form,

$$\theta^x (1 - \theta)^{n-x} = \theta^{(x+1)-1} (1 - \theta)^{(n-x+1)-1},$$

which is precisely the shape of a $\text{Beta}(x+1, n-x+1)$ density (B.6.2). A density's shape determines it (the constant is forced by total mass 1), so

$$\theta | X = x \sim \text{Beta}(x+1, n-x+1),$$

and the normalizing integral $m(x)$ of B.4.3 was never touched. By B.6.3, the Bayes estimate—the posterior mean—is

$$a^*(x) = E\{\theta | X = x\} = \frac{x+1}{(x+1) + (n-x+1)} = \frac{x+1}{n+2}.$$

Example B.7.2

The trial's numbers

With $n = 10$ and $x = 7$: the posterior is $\text{Beta}(8, 4)$, and the Bayes estimate is

$$a^*(7) = \frac{7+1}{10+2} = \frac{8}{12} = \frac{2}{3} \approx 0.667,$$

against the raw rate's $x/n = 0.700$. The posterior also prices its own uncertainty: the $\text{Beta}(\alpha, \beta)$ variance is $\alpha\beta/((\alpha+\beta)^2(\alpha+\beta+1))$,⁷ here

$$\text{Var}(\theta | X = 7) = \frac{8 \cdot 4}{12^2 \cdot 13} = \frac{32}{1872} = \frac{2}{117} \approx 0.0171, \quad \text{sd} \approx 0.131.$$

So the full Bayesian report is: *the cure rate is around 0.67, give or take about 0.13*—an estimate and an honest width, both read off one distribution.

Remark B.7.3

The anatomy of $(x+1)/(n+2)$: a weighted average

Rewrite the Bayes estimate to expose its structure:

$$\frac{x+1}{n+2} = \frac{2 \cdot \frac{1}{2} + n \cdot \frac{x}{n}}{2+n}.$$

(Check: the numerator is $1+x$ and the denominator $2+n$, as before.) Read it as a weighted average of two guesses: the *prior mean* $\frac{1}{2}$, carrying weight 2, and the *data's rate* x/n , carrying weight n . The flat prior behaves exactly like *two phantom patients, one cured and one not*, mixed into the sample.

⁷Derived by the same kernel trick as the mean: $E\{\theta^2\} = B(\alpha+2, \beta)/B(\alpha, \beta) = \frac{(\alpha+1)\alpha}{(\alpha+\beta+1)(\alpha+\beta)}$ (two applications of the recursion), then $\text{Var} = E\{\theta^2\} - E\{\theta\}^2$ with common denominator $(\alpha+\beta)^2(\alpha+\beta+1)$.

The same algebra runs for any $\text{Beta}(\alpha, \beta)$ prior: the posterior is $\text{Beta}(\alpha + x, \beta + n - x)$ —observed successes reinforce α , failures reinforce β —and its mean is

$$\frac{\alpha + x}{\alpha + \beta + n} = \frac{(\alpha + \beta) \cdot \frac{\alpha}{\alpha + \beta} + n \cdot \frac{x}{n}}{(\alpha + \beta) + n} :$$

prior mean weighted by $\alpha + \beta$ (*the prior is worth $\alpha + \beta$ observations*), data rate weighted by n . This is the general shape of Bayesian estimates: a compromise between what you believed and what you saw, at exchange rates set by how much each side knows.

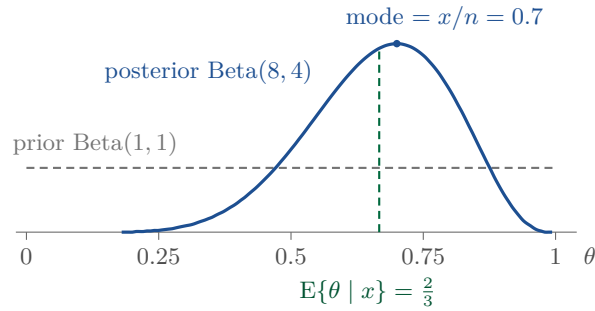


Figure B.7.4. THE UPDATE, DRAWN. The flat prior (dashed) and the posterior $\text{Beta}(8, 4)$ after observing 7 cures in 10 patients. The posterior’s *peak* sits at exactly the raw rate $x/n = 0.7$ (the data’s best single explanation), but its *mean*—the Bayes estimate under squared error, marked in green—sits at $2/3 \approx 0.667$, pulled toward the middle by the distribution’s left tail. The whole curve, not just one number, is the updated state of knowledge; the loss function then selects which one number to report (B.5.1).

Remark B.7.5

Two sanity checks

No data. Set $n = x = 0$: the estimate is $(0 + 1)/(0 + 2) = \frac{1}{2}$, the prior mean—with nothing observed, the recipe reports what you believed. *All cures.* Set $x = n$: the estimate is $(n + 1)/(n + 2)$, which is large but strictly less than 1: ten out of ten cures should not convince anyone that failure is *impossible*. (The raw rate $x/n = 1$ would assert exactly that.) This estimator for the all-successes case is old and famous—it is *Laplace’s rule of succession*, his answer to “what probability that the sun rises tomorrow, given that it always has?”—and it falls out of the machinery here as a routine special case.

B.8 A Second Instrument: the Normal–Normal Pair

The Beta–binomial pair handled a fraction. The other workhorse setting is a *measurement*: a quantity on the whole line, observed through noise. One new distribution is needed.

Definition B.8.1

The normal distribution

The *normal* (or Gaussian) distribution $N(\mu, \sigma^2)$ is the continuous distribution with density

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right), \quad y \in \mathbb{R} :$$

the bell curve, centered at its mean μ , with spread set by its standard deviation σ (about 68% of the probability lies within one σ of μ , about 95% within two). Its ubiquity has a theorem behind it (sums of many small independent effects are approximately normal—the central limit theorem, not needed here) and a practical reason: as a function of the variable in the exponent, the density is $e^{-\text{quadratic}}$, and quadratics are closed under addition. That closure is exactly what the next example exploits.

Example B.8.2

Normal prior, normal measurement: the posterior in full

Setup. A quantity θ (an aptitude score, a physical constant, a true weight) has prior $\theta \sim N(m, \tau^2)$: best prior guess m , prior uncertainty τ . An instrument measures it once with known noise level σ : given the truth, $X | \theta \sim N(\theta, \sigma^2)$. Observe $X = x$. What is the posterior? *Multiply prior by likelihood* (B.4.5), keeping only factors involving θ —the two normalizing constants $1/(\tau\sqrt{2\pi})$ and $1/(\sigma\sqrt{2\pi})$ drop:

$$\pi(\theta | x) \propto \exp\left(-\frac{(\theta - m)^2}{2\tau^2}\right) \cdot \exp\left(-\frac{(x - \theta)^2}{2\sigma^2}\right) = \exp\left(-\frac{1}{2} \left[\frac{(\theta - m)^2}{\tau^2} + \frac{(\theta - x)^2}{\sigma^2} \right]\right).$$

Expand the bracket and collect powers of θ . Expanding each square,

$$\frac{\theta^2 - 2m\theta + m^2}{\tau^2} + \frac{\theta^2 - 2x\theta + x^2}{\sigma^2} = \theta^2 \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2} \right) - 2\theta \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) + \left(\frac{m^2}{\tau^2} + \frac{x^2}{\sigma^2} \right),$$

and the last group is free of θ —bound for the normalizing constant. *Complete the square.* Abbreviate

$$\frac{1}{v} = \frac{1}{\tau^2} + \frac{1}{\sigma^2}, \quad \frac{m^*}{v} = \frac{m}{\tau^2} + \frac{x}{\sigma^2} \quad (\text{defining } v \text{ and } m^*),$$

so the θ -dependent part is

$$\frac{\theta^2 - 2m^*\theta}{v} = \frac{(\theta - m^*)^2 - (m^*)^2}{v},$$

by adding and subtracting $(m^*)^2$ inside—and $(m^*)^2/v$, constant in θ , also departs for the normalizer. What remains is

$$\pi(\theta | x) \propto \exp\left(-\frac{(\theta - m^*)^2}{2v}\right) :$$

the kernel of a normal density (B.8.1). Matching shapes,

$$\theta | X = x \sim N(m^*, v), \quad m^* = v \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) = \frac{\sigma^2 m + \tau^2 x}{\sigma^2 + \tau^2},$$

the last simplification by multiplying numerator and denominator by $\sigma^2\tau^2$:

$$v \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) = \frac{\frac{m}{\tau^2} + \frac{x}{\sigma^2}}{\frac{1}{\tau^2} + \frac{1}{\sigma^2}} = \frac{\sigma^2 m + \tau^2 x}{\sigma^2 + \tau^2}.$$

By B.5.4 the Bayes estimate is m^* , and the posterior variance v prices the remaining uncertainty.

Example B.8.3

Numbers: one loud measurement

Prior: $\theta \sim N(100, 15^2)$ (say, an aptitude score known to average 100 with spread 15 in the population the subject comes from). Instrument: $\sigma = 10$. Observation: $x = 130$ —a strikingly high reading. Posterior, by B.8.2, in exact fractions:

$$\frac{1}{v} = \frac{1}{225} + \frac{1}{100} = \frac{4}{900} + \frac{9}{900} = \frac{13}{900} \quad \Rightarrow \quad v = \frac{900}{13} \approx 69.2, \quad \sqrt{v} = \frac{30}{\sqrt{13}} \approx 8.3;$$

$$m^* = v \left(\frac{100}{225} + \frac{130}{100} \right) = \frac{900}{13} \left(\frac{4}{9} + \frac{13}{10} \right) = \frac{900}{13} \cdot \frac{40 + 117}{90} = \frac{10 \cdot 157}{13} = \frac{1570}{13} \approx 120.8.$$

The Bayes estimate is about 120.8: well above the prior's 100 (the measurement was loud and the instrument decent), but pulled down meaningfully from the raw reading 130 (a single reading, from a population where 130 is rare, is partly signal and partly luck). The posterior spread 8.3 is smaller than *both* the prior's 15 and the instrument's 10: two imperfect sources of information, combined, know more than either alone.

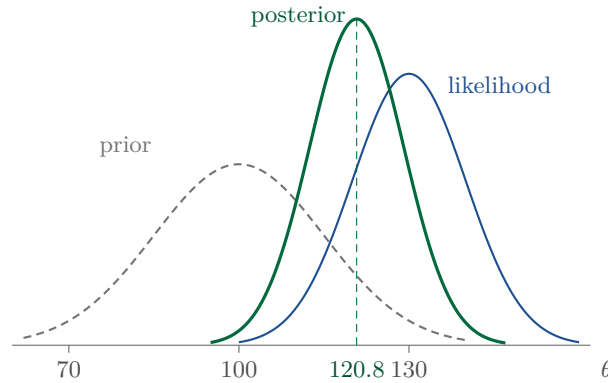


Figure B.8.4. PRIOR, LIKELIHOOD, POSTERIOR. The update of B.8.3: prior $N(100, 15^2)$ (dashed), the likelihood of the observation $x = 130$ (blue, a normal shape in θ centered at the reading), and the posterior $N(120.8, 8.3^2)$ (green). The posterior sits between its two parents, closer to the likelihood because the instrument ($\sigma = 10$) is more precise than the prior ($\tau = 15$)—and it is narrower than either, because precisions add (B.8.5). Curves drawn to a common vertical scale.

Remark B.8.5

The precision reading, and shrinkage

State B.8.2's formulas in terms of *precision* = $1/\text{variance}$ (how much a source of information knows):

$$\underbrace{\frac{1}{v}}_{\text{posterior precision}} = \underbrace{\frac{1}{\tau^2}}_{\text{prior precision}} + \underbrace{\frac{1}{\sigma^2}}_{\text{data precision}}, \quad m^* = \frac{\frac{1}{\tau^2}}{\frac{1}{\tau^2} + \frac{1}{\sigma^2}} m + \frac{\frac{1}{\sigma^2}}{\frac{1}{\tau^2} + \frac{1}{\sigma^2}} x :$$

precisions add, and the posterior mean weighs each guess by its share of the total precision—the continuous twin of B.7.3’s observation-counting. In the numbers of B.8.3 the weights are $\frac{4}{13}$ on the prior and $\frac{9}{13}$ on the data: $m^* = \frac{4}{13} \cdot 100 + \frac{9}{13} \cdot 130 = \frac{400+1170}{13} = \frac{1570}{13}$, as before. Equivalently,

$$m^* = m + \frac{\tau^2}{\sigma^2 + \tau^2} (x - m) = 100 + \frac{225}{325} (130 - 100) = 100 + \frac{9}{13} \cdot 30 \approx 120.8 :$$

the estimate starts at the prior mean and covers only the fraction $\tau^2/(\sigma^2 + \tau^2)$ of the distance to the data. Bayesian estimates *shrink* noisy observations toward what was already believed—gently when the data are precise, firmly when they are not. This one word, *shrinkage*, is the modern doorway from this handout into a large part of contemporary statistics and machine learning.

B.9 What to Make of the Prior

Remark B.9.1

The estimate depends on the prior—as it must

Rerun the trial of B.7.2 under a different premise. Suppose historical experience with similar treatments made low cure rates plausible—say a Beta(2,6) prior, mean $2/8 = 0.25$, worth $\alpha + \beta = 8$ phantom observations. By B.7.3 the posterior after 7 cures in 10 is Beta($2 + 7, 6 + 3$) = Beta(9,9), with mean

$$\frac{2 + 7}{2 + 6 + 10} = \frac{9}{18} = 0.500,$$

against 0.667 under the flat prior and 0.700 raw. Ten patients cannot overrule eight phantoms. This is not a defect; it is the premise doing exactly what it claimed: with little data, background knowledge *should* matter, and the machinery is transparent about how much. But it lays an obligation on the analyst—the prior is part of the argument, must be stated, and must be defensible on grounds *outside* the data being analyzed. Estimating the prior’s location from the same data it will then shrink is circular, and the circle does not close.⁸

Remark B.9.2

The data win in the end

The dependence on the prior is a small-sample phenomenon. Compare the Bayes estimate with the observed rate at stage n by putting them over a common denominator:

$$\frac{\alpha + x}{\alpha + \beta + n} - \frac{x}{n} = \frac{n(\alpha + x) - x(\alpha + \beta + n)}{n(\alpha + \beta + n)} = \frac{\alpha n - x(\alpha + \beta)}{n(\alpha + \beta + n)} = \frac{\alpha - (\alpha + \beta) \frac{x}{n}}{\alpha + \beta + n},$$

the middle step cancelling the xn terms. Since the rate x/n lies in $[0, 1]$, the numerator can be no larger in size than $\max(\alpha, \beta)$ —a fixed number—while the denominator grows without bound. So the gap between the Bayes estimate and the observed rate shrinks to zero as data accumulate, *whatever* the data do: every fixed Beta prior’s estimate is eventually indistinguishable from the raw rate—the phantoms are outvoted. Priors differing in judgment but agreed on possibilities are eventually reconciled by evidence; where the prior genuinely matters is exactly where matters are genuinely uncertain: small samples.

⁸The course deck’s aphorism, quoted in the notes at 2.10.2: “you cannot use the data to know what to shrink towards; but you can use the data to know how much to shrink towards it”—the *amount* of shrinkage, unlike the *target*, is learnable when many parallel problems share one prior.

Remark B.9.3*Two schools, one course*

A statistician who declines to put a distribution on θ —who insists the cure rate is a fixed fact, not a random variable—works instead with the risk *curves* of B.3.4 and principles for comparing them (ruling out estimators beaten everywhere; guarding the worst case). That is the *frequentist* school, and it is where the course notes spend § 1.5. The Bayesian school, this handout’s subject, buys a complete and unique answer at the price of a prior. The two are not enemies so much as different contracts with the reader about what may be assumed—and they meet constantly: the course’s Lecture 1 appendix shows a Bayesian-derived rule, $(x+1)/(n+2)$, holding its own in a purely frequentist risk comparison (1.A.4, 1.A.5). A working statistician reads both languages.

B.10 The Vocabulary, and Where to Go Next

Concept B.10.1*The names of everything you just did*

For searching, reading, and talking to statisticians—the standard names, each pointing at its object in this handout:

- *Prior distribution* (B.4.1); *likelihood* (B.4.5); *posterior distribution* (B.4.4); *Bayes’ theorem* (B.4.4); *Bayesian updating* (the whole § B.4 pipeline).
- *Bayes risk* (B.5.3); *Bayes estimator* or *Bayes rule* (B.5.4)—beware that “Bayes’ rule” also names the updating theorem; context decides. The posterior-mean result is often quoted as “the Bayes estimator under squared-error (quadratic) loss is the posterior mean.”
- *Conjugate prior*: a family, like Beta for the binomial (§ B.7) or normal for the normal (§ B.8), whose posteriors stay in the family—the kernel trick succeeds by design. *Shrinkage* (B.8.5); *Laplace’s rule of succession* (B.7.5).
- In engineering literature the § B.5 mathematics runs under *minimum mean squared error (MMSE) estimation*: the conditional expectation as the MMSE predictor.
- One report this handout did not build: an *interval* from the posterior (e.g., the central 95% of Beta(8,4)), called a *credible interval*—the Bayesian counterpart of the confidence sets in the course notes, § 2.7.

Concept B.10.2*Where to go next*

In the course notes: § 2.8 proves the two prediction lemmas in their general form, § 2.9 is this handout’s material at course pace (with the Poisson–gamma pair as a third conjugate example), and 1.A.5 runs the trial’s computation inside the decision-theory frame of Lecture 1. In the texts: Hoel–Port–Stone treat Bayesian methods in § 2.8 (their Theorem 2 is B.5.4); DeGroot & Schervish devote Chapter 7 (§§ 7.2–7.4) to priors, posteriors, and Bayes estimators at a gentle pace with many worked examples; Blitzstein & Hwang’s Chapters 7–9 are the friendliest place to firm up the conditional-probability foundations of § B.2. And carry out one exercise by hand before reading further: redo § B.7 with your own prior—pick α and β to express something you actually believe about some fraction in your life, invent data, and push it through B.7.3. The method fits on an index card; the point of the theory was to earn it.

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